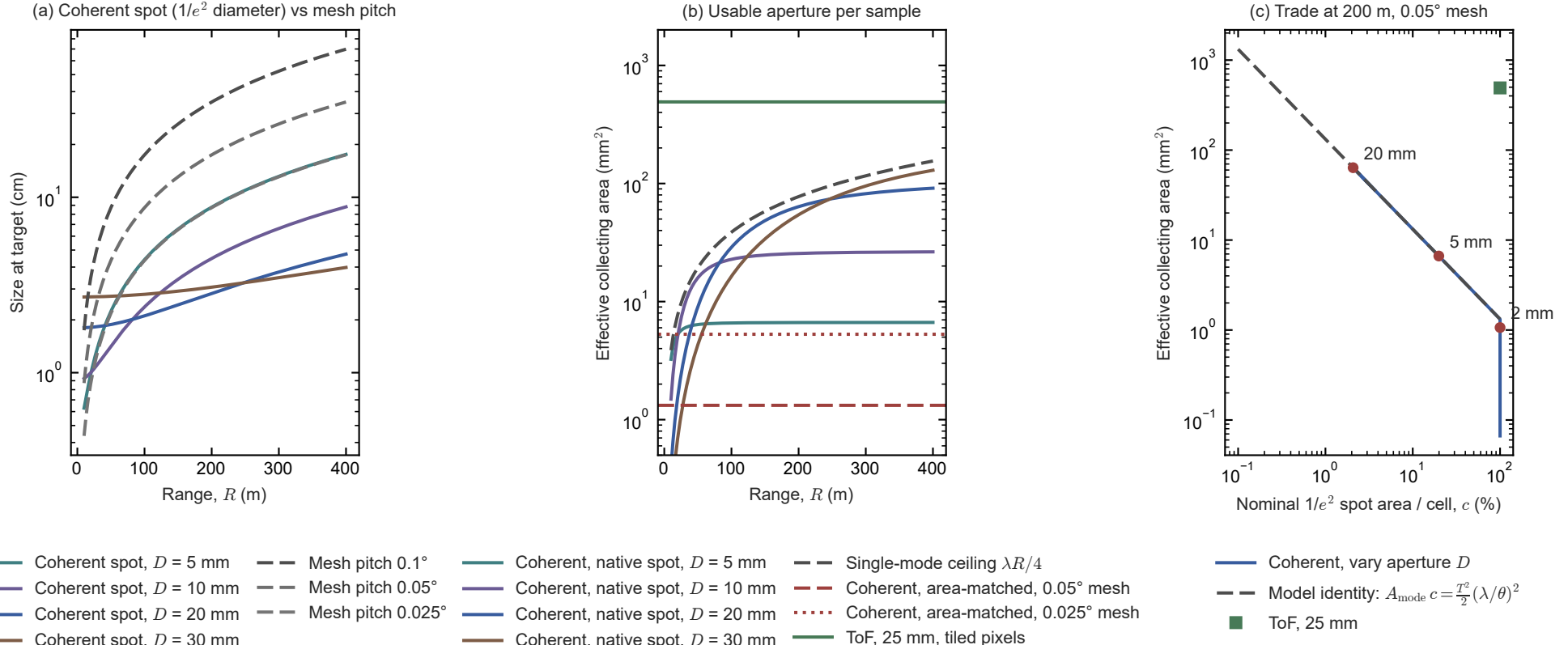
(a) Coherent spot ($1/e^2$ diameter) vs mesh pitch
Size at target (cm)
Range, R (m)
(b) Usable aperture per sample
Effective collecting area (mm^2)
Range, R (m)
(c) Trade at 200 m, 0.05° mesh
Effective collecting area (mm^2)
20 mm
5 mm
2 mm
Nominal $1/e^2$ spot area / cell, c (%)
Coherent spot, D = 5 mm
Coherent spot, D = 10 mm
Coherent spot, D = 20 mm
Coherent spot, D = 30 mm
Mesh pitch 0.1°
Mesh pitch 0.05°
Mesh pitch 0.025°
Coherent, native spot, D = 5 mm
Coherent, native spot, D = 10 mm
Coherent, native spot, D = 20 mm
Coherent, native spot, D = 30 mm
Single-mode ceiling $\lambda R/4$
Coherent, area-matched, 0.05° mesh
Coherent, area-matched, 0.025° mesh
ToF, 25 mm, tiled pixels
Coherent, vary aperture D
Model identity: $A_{\mathrm{mode}}\, c = \frac{\pi^2}{2}(\lambda/\theta)^2$
ToF, 25 mm